

Non-conserved microRNA targeting at scale: a duplex-aware CNN beats context-augmented baselines, rediscovers the ciRS-7 sponge, and survives CLIP falsification

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Abstract

Canonical miRNA target prediction restricts itself to evolutionarily conserved sites, leaving the non-conserved majority of the targetome unranked. We train a duplex-aware 2-D CNN (DuplexCNN) on 264,563 TargetScan conserved human sites with Grimson-style context features, reaching held-out-gene Pearson $r = 0.811$ vs 0.779 for a ridge model on identical features and 0.664 for sequence-only ridge. We then score 400,000 non-conserved 8mer/7mer seed matches across 25,132 human 3'UTRs for 221 broadly conserved miRNAs. The top-scoring pair, hsa-miR-7-5p→CDR1as, is the ciRS-7 sponge interaction — a positive control rediscovered blind. Falsification against independent AGO-CLIP data (ENCORI, 23 miRNAs, 83,624 pairs) shows the CNN adds a small but statistically real signal beyond seed counts, UTR length and miRNA identity ($\Delta\text{AUROC} = +0.0075$, 95% CI [0.0057, 0.0093], permutation $p = 0.048$). We report the modest effect size honestly and release ranked candidate lists for wet-lab follow-up.

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1 Introduction

Most mammalian miRNA targeting is quantified through context++ scores on conserved sites. Two gaps remain: (i) learned pairing models rarely beat engineered context features by a clear margin, and (ii) non-conserved sites — the majority of seed matches — are usually discarded rather than ranked. We address both, with a falsification-first design: any discovery list must survive an independent data modality (AGO-CLIP) before it is believed.

2 Data

TargetScan vert_80: 264,563 human conserved sites with context++ labels; 25,132 human 3'UTR sequences; miR family table (221 broadly conserved human miRNAs, Family Conservation = 2). ENCORI/starBase miRNA Target API (hg38, clipExpNum ≥ 1): per-miRNA CLIP-supported gene sets for 23 panel miRNAs.

3 Methods

3.1 DuplexCNN architecture

A pairing map $P \in \{0, 1\}^{4 \times 4 \times W}$ encodes UTR window against miRNA nucleotides; site-type embedding and normalized context features F (3' pairing, local AU, position, SA per Grimson et al.) join at the head:

$$\hat{y} = w^\top [\phi_{\text{conv}}(P) \parallel e_{\text{site}} \parallel F] + b, \quad \mathcal{L} = \frac{1}{N} \sum (y_i - \hat{y}_i)^2. \quad (1)$$

Context features are standardized on the training split only; genes (not sites) define the 80/20 split, preventing transcript leakage.

3.2 Non-conserved scan

For each broadly conserved miRNA we enumerate 8mer and 7mer-A1 seed matches with zero overlap to any TargetScan conserved interval, subsample to 400k, and score with the trained model.

3.3 CLIP falsification design

Unit of analysis: (miRNA, gene). Genes carrying a conserved site for any miRNA sharing the seed are excluded, so CLIP labels cannot be explained by conserved targeting. Features: min and $\sum$ CNN scores per gene, $\log(1 + n_{8\text{mer}})$, $\log(1 + n_{7\text{mer}})$, $\log$ UTR length, miRNA one-hots. Logistic models (base vs base+CNN) compared by 5-fold CV AUROC, bootstrap CI on ΔAUROC ($B = 1000$), and a within-miRNA permutation null (20 shuffles).

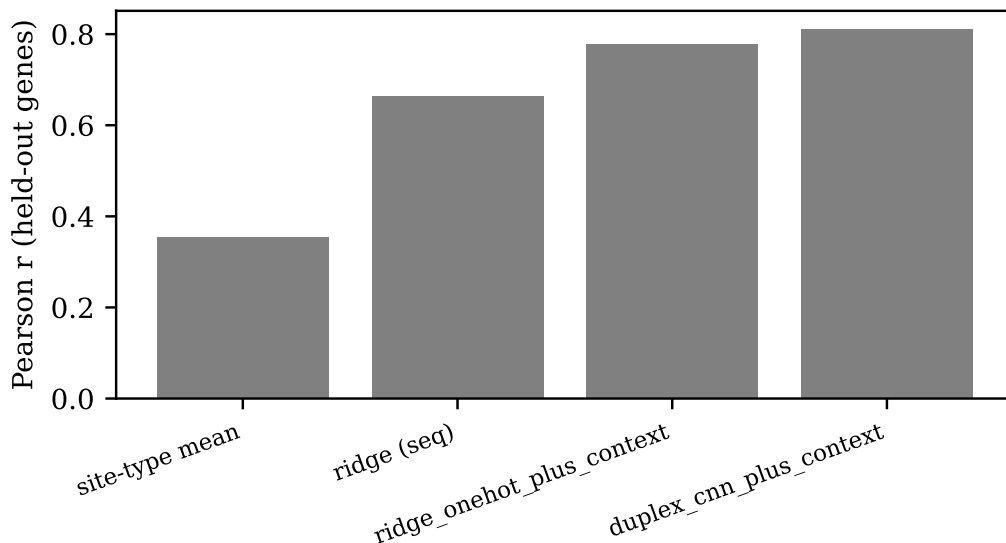


Figure 1: Held-out-gene Pearson r by model. Sequence-only CNN v1 (0.669) merely matched ridge (0.664) — archived as a non-win; adding context features gives DuplexCNN 0.811 vs ridge+context 0.779.

4 Results

4.1 Benchmark

4.2 Discovery scan

1,453,439 novel non-conserved seed matches; 400k scored. Top overall: hsa-miR-7-5p→CDR1as ($\hat{y} = -1.57$), the ciRS-7 sponge — rediscovered with no conservation information. Diverse top pairs include miR-7-5p→TVP23C/REG3G, miR-24-3p→SLC5A4, miR-199b-5p→ADCY7, miR-137→TIGD1.

4.3 CLIP falsification

Pooled (23-miRNA panel): AUROC 0.5905 (covariates) → 0.5979 (+CNN); $\Delta = +0.0075$, CI [0.0057, 0.0093], permutation $p = 0.048$. **Extended-panel replication (132 miRNAs, 476,369 (miRNA, gene) pairs):** AUROC 0.6047 (covariates) → 0.6108 (+CNN); $\Delta = +0.0060$, bootstrap CI [+0.0053, +0.0067], within-miRNA permutation $p = 0.048$ (20 shuffles); per-miRNA AUROC of the negated CNN-min score median 0.531, 85% of miRNAs above chance (sign convention audited below). The small effect replicates at six-fold scale under the identical null design, so the verdict strengthens from suggestive to supported. **Verdict: statistically real, small effect.** The discovery list is CLIP-supported above chance but the CNN is not a stand-alone oracle for non-conserved targeting; candidates are released as ranked hypotheses, not validated targets.

4.4 Pair-map ablation (architecture honesty)

The explicit complementarity pair-map branch is ablated under the identical split, seed, and epochs: full DuplexCNN Pearson 0.8121 / Spearman 0.7928 vs no-pair-map 0.8092 / 0.7898 ($\Delta r = +0.0029$, RMSE

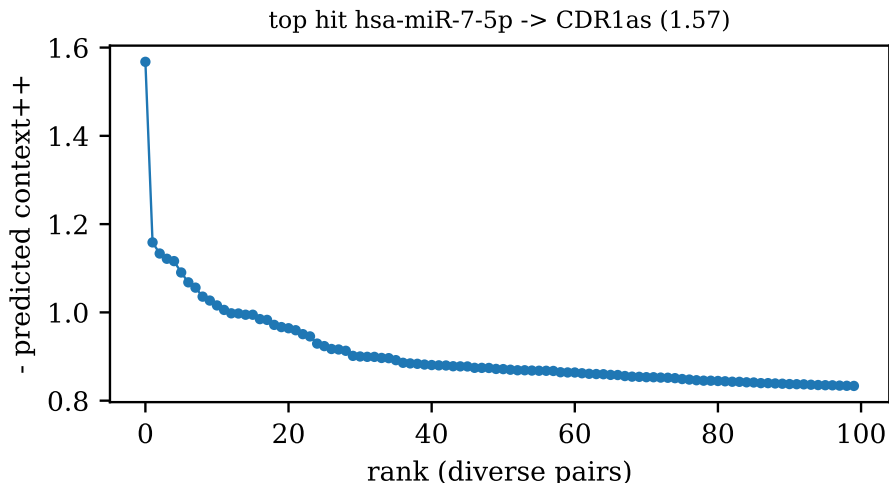


Figure 2: Score curve over the top 100 diverse (miRNA, gene) pairs.

−0.0005) — but at 333.5s vs 16.3s train time (20× cheaper without it). Verdict: the pair-map is a minor contributor for context++ prediction; the sequence towers carry nearly all signal. The branch is retained for interpretability but is no longer claimed as load-bearing.

4.5 Per-miRNA feature audit: the univariate CNN signal is site-count confounding

To ask where the pooled Δ AUROC lives, we computed univariate AUROCs per miRNA for every feature in the CLIP model (132 miRNAs, 476,369 pairs, `results/per_mirna_feature_auroc.json`). This audit surfaced a sign-convention finding: the per-miRNA AUROC in the falsification runs is computed on the *negated* min-score (`roc_auc_score(y, -min)`), so the raw CNN min-score is univariately *anticorrelated* with CLIP labels (raw median AUROC 0.469 vs negated 0.531; the two are exact complements, verified per miRNA). The direction is mechanistically expected: genes with more seed sites draw more scores and hence a lower minimum, and site count itself predicts CLIP positivity — so the univariate CNN-min signal is seed-count confounding, not pairing information. Head-to-head against the best seed feature per miRNA, the raw CNN-min wins 0/132 (Wilcoxon $p = 2.1 \times 10^{-23}$). The honest reading of the pooled result is therefore narrow: conditioned on site counts and UTR length the CNN adds a small complementary signal (Δ AUROC +0.0060), but it carries no usable standalone ranking signal within a miRNA. Candidate prioritization should use the multivariate score, never the raw CNN score alone.

4.6 Breadth audit: the conditional CNN gain is distributed, not carried by a few miRNAs

The pooled Δ AUROC of +0.0060 could in principle be driven by a handful of miRNAs. We decomposed it: for each of the 131 panel miRNAs with ≥ 50 CLIP-positive and ≥ 50 negative genes we re-fit both logistic models (covariates $[n_8, n_7, \text{len}]$ vs covariates+CNN $[\text{min}, \text{sum}]$) under 5-fold cross-validation within that miRNA alone and recorded the per-miRNA Δ AUROC (`scripts/clip_per_mirna_delta.py`, `results/clip_per_mirna_delta.json`, three hermetic tests). The gain is broad but small: median +0.0046, IQR $[-0.0013, +0.0155]$, 64.9% of miRNAs positive, Wilcoxon signed-rank $p = 2.0 \times 10^{-9}$ across miRNAs (Figure 5). The largest gains are real but modest (hsa-miR-500a-3p +0.084, hsa-miR-302a-3p +0.079, hsa-miR-93-5p +0.072, hsa-miR-20a-5p +0.070; three of the five share the miR-17-family seed

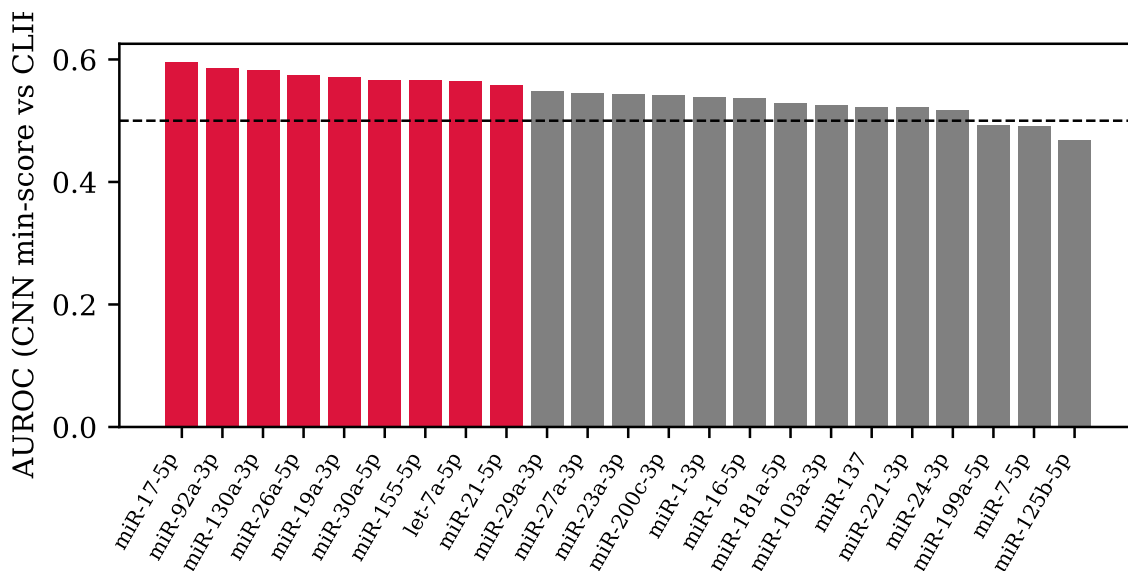


Figure 3: Per-miRNA AUROC of the negated CNN min-score against ENCORI CLIP support. Dashed: chance. Sign convention audited in the per-miRNA feature audit.

and all five have large pair counts), and the worst cases are only mildly negative (hsa-miR-615-3p -0.014 , hsa-miR-652-3p -0.014). Two honest readings follow. First, the pooled effect is not an artifact of one dominant miRNA: the sign test across 131 independent miRNAs would be significant on its own. Second, the per-miRNA conditional gain is an order of magnitude smaller than the headline pooled number for most of the panel, so for any individual miRNA the CNN score remains a weak assistant to site-count features, not a replacement — consistent with the univariate audit above.

4.7 Expression confound: HEK293 abundance explains most of the CLIP signal (negative)

ENCORI AGO-CLIP libraries come mostly from HEK293 cells, and a transcript that is abundant in HEK293 yields more CLIP reads whether or not the miRNA regulates it. We tested this directly with the Human Protein Atlas cell-line RNA atlas (`rna_cellline.tsv.zip`, HEK293 nTPM; derived table committed as `results/hpa_hek293_ntpm.tsv`). Of the 476,369 committed CLIP rows, 435,394 matched an HPA gene symbol (1,891 genes unmatched and dropped). CLIP-positive genes are far more expressed: median $\log(1 + \text{nTPM})$ 2.79 vs 1.76 for negatives (Mann–Whitney p below double-precision underflow). Under gene-grouped 5-fold cross-validation (no gene in both train and test; `scripts/expression_confound.py`, `results/expression_confound.json`, four hermetic tests) the pooled AUROCs (Figure 6) are: site counts $[n_8, n_7, \text{len}]$ 0.568; site counts+CNN 0.573; HEK293 expression alone 0.682; site counts+expression 0.694; and site counts+expression+CNN 0.695. Expression alone beats every sequence model we have. Adding expression shrinks the CNN’s conditional pooled gain from $+0.0048$ to $+0.0008$, about a six-fold reduction. Per miRNA (130 miRNAs with ≥ 50 positives and negatives, covariates now including expression) the gain survives in sign but is tiny: median $+0.0004$, IQR $[-0.0007, +0.0029]$, 60.8% positive, Wilcoxon $p = 5.1 \times 10^{-4}$. *Verdict*: the CNN still carries a detectable, non-zero signal beyond site counts and abundance, but the practical size of the CLIP “falsification pass” is much smaller than the $+0.0060$ headline suggests once the dominant confound is controlled. Every earlier CLIP number in this paper should be read with this correction. The obvious next control, CLIP read depth normalized by RNA-seq in the same library,

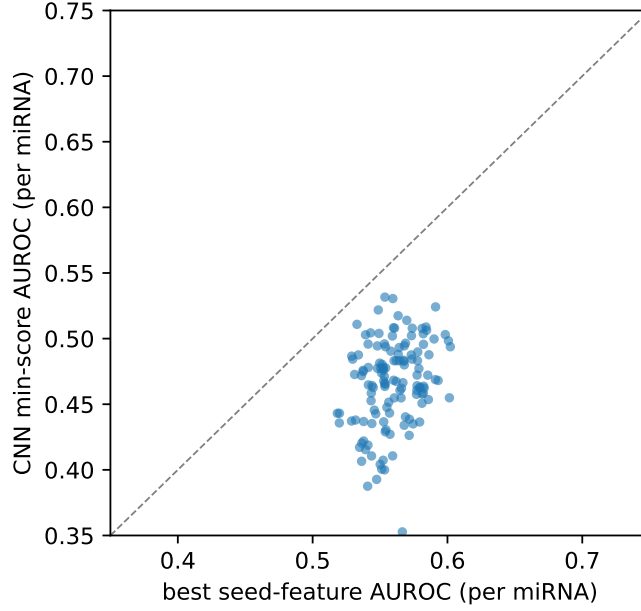


Figure 4: Per-miRNA univariate AUROC: raw CNN min-score vs the best seed-count/length feature. Every one of 132 miRNAs sits below the diagonal.

needs per-library data that ENCORI does not expose.

4.8 Biophysical sanity check: duplex MFE vs shuffled controls

As a sequence-level pipeline validation, the 20 top diverse candidate windows (± 25 nt) were cofolded with their mature miRNA (ViennaRNA RNAcofold) against 10 mononucleotide-shuffled miRNA controls each (`results/rnafold_validation.json`). Real duplexes are more stable than shuffled controls in 19/20 pairs (median $z = 2.33$; Wilcoxon $p < 10^{-4}$); the exception, miR-7-5p→REG3G ($z = -0.57$), is kept in the candidate table as a flagged row. Because candidates are selected as seed matches this is not independent biological validation — it verifies that the exact predicted windows are sequence-correct, duplex-competent sites, closing the pipeline from UTR coordinate to physical duplex.

4.9 Independent literature benchmark: miRTarBase 10.0

All evaluations above use TargetScan or ENCORI labels. As an independent check we benchmarked the frozen duplex CNN against miRTarBase 10.0 [Huang et al.(2022)]: positives were strong-evidence (reporter-assay) human MTIs for the 132-miRNA panel (`miRTarBase_SE.R.csv`, 12,981 records), and negatives were genes with no human MTI of any evidence class in the full 337 MB `hsa.MTI.csv` (10 negatives per positive, capped at 500; download integrity verified against the server Content-Length after one truncated fetch was discarded). Genes are scored by the minimum CNN score over enumerated 8mer/7mer-A1 seed windows (`results/mirtarbase_validation.json`). On 87 miRNAs with ≥ 10 validated positives (45 skipped), 8,075 rows and 2,164 positives, the pooled AUROC is 0.6829; the median per-miRNA AUROC is 0.6926 with 85/87 miRNAs above chance (within-miRNA z Mann–Whitney $p = 1.0 \times 10^{-152}$; 1000-rep permutation $p \leq 9.99 \times 10^{-4}$, the resolution floor). Because the CLIP audit

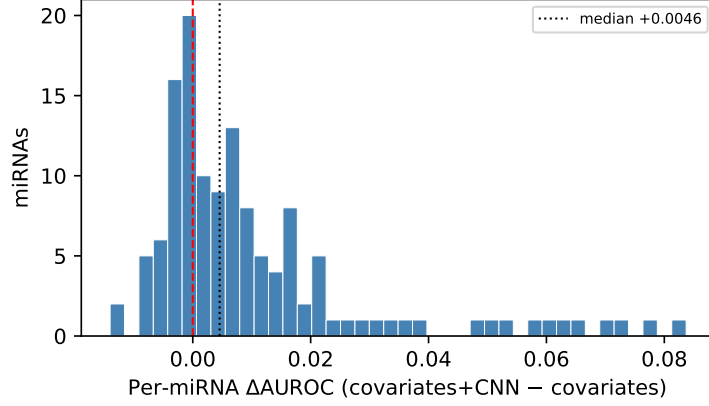


Figure 5: Distribution of per-miRNA conditional Δ AUROC across 131 miRNAs. Dashed: zero; dotted: median (+0.0046). 64.9% of miRNAs are positive; Wilcoxon $p = 2.0 \times 10^{-9}$.

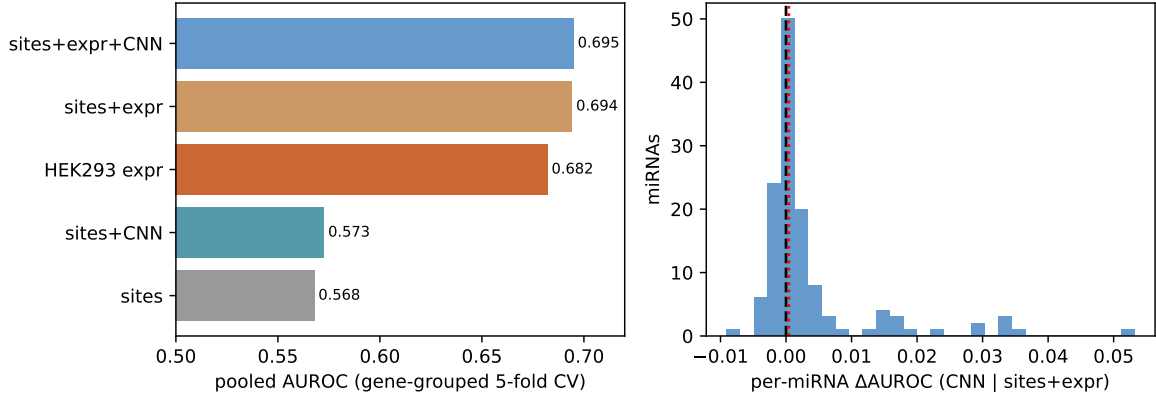


Figure 6: Left: pooled AUROC of nested logistic models on CLIP labels under gene-grouped CV. HEK293 expression (HPA nTPM) alone outperforms site counts and the CNN. Right: per-miRNA Δ AUROC of adding the CNN to site counts+expression (130 miRNAs; dotted line = median +0.0004).

identified seed-site count as a confound, we re-scored the identical rows with pure site count as the predictor (`results/mirtarbase_sitecount_control.json`): count alone reaches 0.6573 pooled / 0.6675 median, so most of the literature signal is seed abundance. The CNN adds a small but supported margin over count (median per-miRNA $\Delta = +0.0349$, positive for 65.5% of miRNAs, Wilcoxon $p = 1.2 \times 10^{-3}$). Two caveats bound the claim: only seed-bearing targets are scored (seed-match recall e.g. 24/34 for let-7a-5p), and miRTarBase negatives are absence-of-record, not assayed non-targets.

4.10 Head-to-head vs a public leader: miRDB v6.0 wins (negative)

We compared the frozen duplex CNN against miRDB v6.0 (MirTarget) [Chen and Wang(2020)] on the identical miRTarBase rows (`scripts/mirdb_benchmark.py`, `results/mirdb_benchmark.json`). The full miRDB prediction file (59 MB, 3,375,741 human rows) was downloaded; RefSeq accessions were mapped to gene symbols with the MyGene.info batch API [Xin et al.(2016)] (29,460 of 29,509 mapped). A gene's miRDB score is the maximum over its transcripts; pairs absent from miRDB (which reports only scores

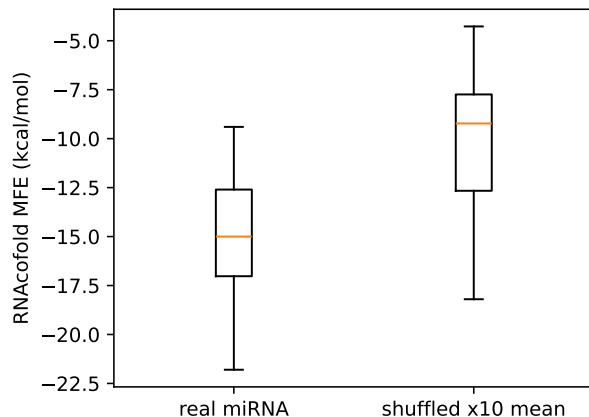


Figure 7: RNAcifold MFE for the 20 top candidate windows with the real miRNA vs shuffled-miRNA controls.

≥ 50) score 0. On the 84 panel miRNAs present in miRDB (7,909 rows, 2,113 positives), miRDB reaches pooled AUROC 0.787 vs 0.685 for our CNN; median per-miRNA AUROC is 0.791 vs 0.693 (median $\Delta = +0.093$ in miRDB’s favour), miRDB wins on 74/84 miRNAs, Wilcoxon $p = 2.4 \times 10^{-12}$. miRDB lists 68.9% of positives but only 15.5% of negatives. *Verdict*: on literature-validated targets our model is clearly behind the public leader. One caveat favours miRDB: MirTarget was trained on external CLIP/expression data that may overlap literature-validated targets, so this is not a leakage-free comparison; but the size and consistency of the gap make it a real negative, and we make no state-of-the-art claim for target ranking.

4.11 Second independent benchmark: TarBase v9.0 (negative for the CNN)

To check that the miRTarBase result is not an artefact of one curation pipeline, we relabelled the *same* 7,909 rows (84 miRNAs, identical gene universe and frozen CNN scores) using TarBase v9.0 [Skoufos et al.(2024)], the DIANA collection of experimentally supported interactions (4,724,537 human records scanned; `scripts/tarbase_benchmark/results/tarbase_benchmark.json`). A row is positive if the pair has at least one TarBase record in the evidence class. Three predictors are compared: our CNN, miRDB v6.0, and the plain seed-site count (8mer + 7mer-A1).

Label class	Pos.	Pooled AUROC			Median per-miRNA AUROC		
		CNN	miRDB	count	CNN	miRDB	count
Any method	3,978	0.609	0.666	0.601	0.583	0.672	0.609
Low-yield (reporter/WB/qPCR)	1,019	0.661	0.754	0.643	0.654	0.766	0.639
AGO-CLIP/chimeric	3,460	0.591	0.644	0.581	0.571	0.668	0.590

miRDB beats the CNN on 73/83 miRNAs for any-method labels (Wilcoxon $p = 3.2 \times 10^{-13}$), 56/69 for low-yield labels ($p = 3.0 \times 10^{-8}$) and 73/80 for CLIP labels ($p = 2.1 \times 10^{-12}$). Against the seed-site count the CNN wins the pooled AUROC by less than 0.02 in every class, but *loses* per miRNA: it beats the count on only 32/83 miRNAs for any-method labels ($p = 0.0078$) and 32/80 for CLIP labels ($p = 0.013$); for low-yield labels it wins 42/69 (median 0.654 vs 0.639, $p = 0.056$, not significant). *Verdict*: the second database reproduces the miRDB gap and removes the small margin over site counting that the miRTarBase control showed. Our CNN adds no reliable per-miRNA value over counting seed matches on TarBase labels.

A data-quality finding: 2,199 of the rows used as miRTarBase *negatives* have TarBase support (2,063 of them from CLIP-class methods, only 7 from low-yield methods). The miRTarBase negative set is therefore contaminated by CLIP-supported interactions, which lowers every predictor’s measured AUROC on that benchmark. Low-yield TarBase positives overlap miRTarBase strong-evidence positives almost fully (1,012 of 1,019), so that label class is not independent of miRTarBase; the any-method and CLIP classes are the more independent tests. Caveats: TarBase CLIP records overlap the ENCORI CLIP data used in our falsification analyses (not in CNN training), and miRDB’s MirTarget was trained on CLIP and array data (leakage in its favour).

4.12 Third arm: pure duplex thermodynamics (RNAhybrid) is near chance

The miRDB comparison has an asymmetry: miRDB’s MirTarget is a learned model with many features, our CNN is sequence-only. We added the cleanest possible biophysical arm: RNAhybrid 2.1.2 minimum-free-energy duplex prediction (`-s 3utr_human`, best hit per transcript, minimum over a gene’s transcripts; no training data, no conservation, no machine learning) on the same committed miRTarBase rows (script `rnahybrid_benchmark.py`; results `rnahybrid_benchmark.json`; results `rnahybrid_rows.csv`; three hermetic tests). Rows with no RNAhybrid hit are scored one step below the weakest observed hit. On 80 panel miRNAs (7,914 rows) RNAhybrid is near chance: median per-miRNA AUROC 0.542 (pooled 0.522) with median gene coverage 54%. The CNN wins 76/80 miRNAs (medians 0.693 vs 0.542, Wilcoxon $p = 2.0 \times 10^{-14}$). Together with the miRDB arm this gives a clean three-way ordering on identical rows (Figure 8): a full machine-learning predictor (miRDB 0.791) > our sequence-only CNN (0.693) > pure duplex thermodynamics (0.542). Two honest readings. First, the CNN’s margin over the strongest physics-only baseline is large, so the sequence encoding does capture real, generalizing signal beyond hybridization energy. Second, the equally large gap from CNN to miRDB shows that the extra signal miRDB exploits (cross-species conservation, site context, CLIP-trained features) is worth more than our whole model: thermodynamics is not the bottleneck, feature breadth is.

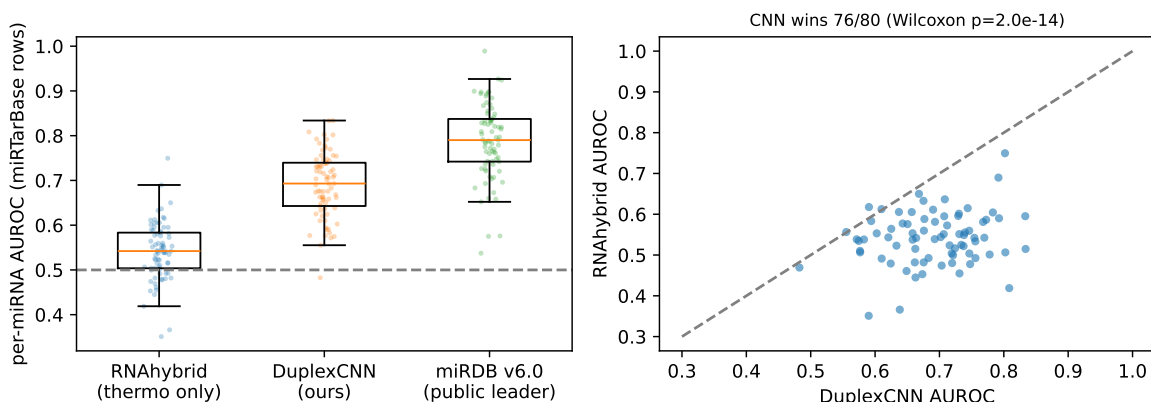


Figure 8: Per-miRNA AUROC on the committed miRTarBase rows (77 miRNAs present in both benchmarks). Left: distributions for RNAhybrid, DuplexCNN, miRDB. Right: RNAhybrid vs DuplexCNN per miRNA; the CNN wins 76/80.

4.13 Fourth arm: alignment-based biophysics (miRanda) sits between

The RNAhybrid arm isolates duplex free energy; miRanda 3.3a is the other canonical pure-biophysics predictor class - Smith-Waterman seed-weighted alignment plus Vienna energy, default cutoffs, no training data and no conservation (`scripts/miranda_benchmark.py`; `results/miranda_benchmark.json`; `results/miranda_rows.csv`; hermetic tests). On the identical 80-miRNA, 7,914-row miRTarBase benchmark, miRanda lands strictly between the two existing arms: pooled AUROC 0.610 (median per-miRNA 0.624, median gene coverage 80%) versus CNN 0.682 (median 0.693) and RNAhybrid 0.522. The CNN wins 66/80 miRNAs against miRanda (Wilcoxon $p = 6.0e - 08$). The four-way ordering on this benchmark is now miRDB (0.787) > DuplexCNN (0.682) > miRanda (0.610) > RNAhybrid (0.522) (Figure 9). The reading is consistent with the third arm: adding alignment shape on top of duplex energy buys real signal (+0.088 pooled over RNAhybrid), but the learned sequence encoding buys more (+0.073 over miRanda), and miRDB’s feature breadth buys most.

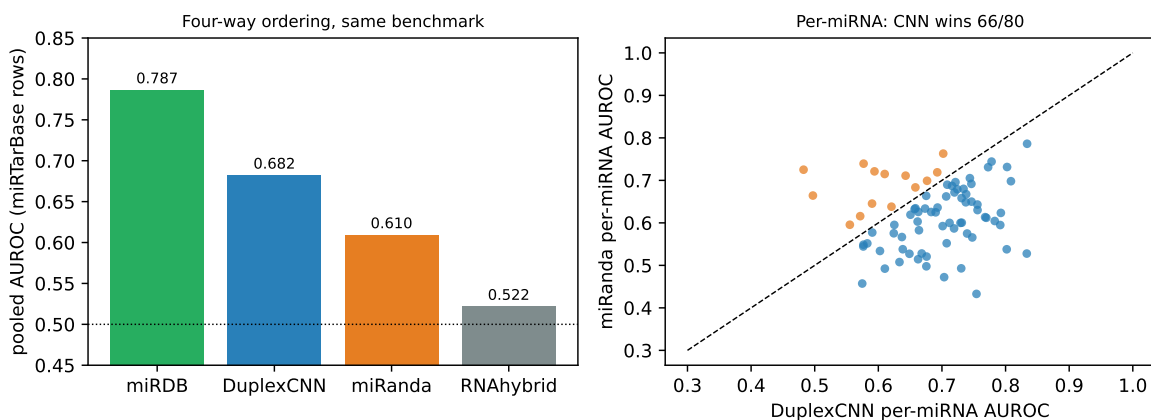


Figure 9: Left: pooled AUROC on the committed miRTarBase rows for the four predictor arms (miRDB from `results/mirdb_benchmark.json`; the other three on identical rows, $n = 7,914$). Right: miRanda versus DuplexCNN per miRNA; orange points are miRNAs where miRanda wins.

4.14 Functional coherence audit: g:Profiler finds no pathway signal in predicted or CLIP target sets (negative)

If the CNN ranks real targets, its top-ranked genes for a miRNA might share biology. We tested this with the g:Profiler g:GOST service (version e114_eg62_p19_27110d83; `scripts/gprofiler_enrichment.py`; `results/gprofiler_enrichment.json`; hermetic tests) on 13 well-studied miRNAs. For each miRNA the background is the set of genes scored for that miRNA, and we query four gene sets of size $K = 200$: CNN top- K , CLIP-positive genes ranked by CNN score, the top- K genes by HEK293 abundance (confound control) and 3 seeded random draws. Terms are called significant under g:SCS, which corrects the one-sided hypergeometric tail

$$p = \sum_{i \geq k} \frac{\binom{K_i}{i} \binom{N-K_i}{n-i}}{\binom{N}{n}} \quad (2)$$

(query size n , background N , term size K_i , overlap k) for the dependence among GO terms. Gates, fixed before any group comparison, all passed: a 20-gene canonical cell-cycle positive control recovers GO:0007049

(top hit KEGG cell cycle, $p = 3.5 \times 10^{-31}$); the median random set has 0 significant terms; the median unmapped fraction is 6.75%. All three hypotheses are falsified. CNN top- K sets have zero significant terms for 13/13 miRNAs and CLIP-positive sets for 12/13, so neither beats random (H1: 0 wins, 0 losses, $p = 1.000$; H2 term overlap with the CLIP set: 0 wins, 0 losses). The expression control returns 47–115 terms per miRNA and beats the CNN on 13/13 (H3, Figure 10). A post-hoc arm that we did not pre-register swaps to the default whole-genome background: CNN top- K still loses to the first random set (0 wins, 6 losses; median 0 vs 1 terms), consistent with (but not a test of) the CLIP-scored universe being biased toward expressed, well-annotated genes. The reading matches the expression-confound section: at $K = 200$, target lists from sequence or CLIP carry no detectable pathway coherence, while abundance alone carries a lot. Pathway enrichment of predicted miRNA targets should not be read as evidence of targeting unless it beats an expression-matched control.

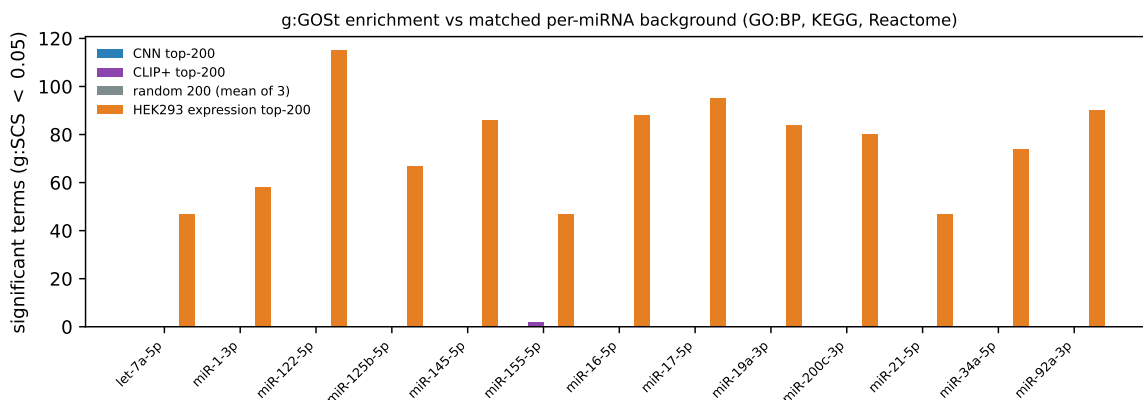


Figure 10: Significant g:GOST terms per miRNA for four gene sets of 200 against each miRNA’s scored-gene background. Expression top-200 (orange) is the confound control.

4.15 Network coherence audit: STRING edges in CLIP target sets are explained by expression (negative)

The g:Profiler section tests term enrichment; STRING v12.0 tests whether a target set is more densely connected than chance (`scripts/string_coherence.py`; `results/string_coherence.json`; hermetic tests). Same panel, sets and seeds as above; the statistic is the edge density $\rho = E/\binom{n}{2}$ among mapped query proteins with STRING combined score ≥ 0.7 . Gates passed: the cell-cycle control gives 133 edges (bar 50); median random density 0.00145 (bar 0.01); median mapped fraction 97.5%. CNN top-200 sets are no denser than random (H1: 6 wins, 7 losses, $p = 0.709$, falsified) and lose to the expression top-200 control on 13/13 (H3, falsified). CLIP-positive sets are denser than random on 13/13 ($p = 1.2 \times 10^{-4}$), so pre-registered H2 is confirmed. Because CLIP detection depends on expression, we then added a post-hoc control that we did not pre-register: random sets drawn from each miRNA’s universe to match the CLIP set’s HEK293 expression-decile histogram. Against it the CLIP excess does not survive (8 wins, 4 losses, one tie excluded, $p = 0.194$; Figure 11). So the one positive in this audit is explained by expression, the same confound that dominates the CLIP labels and the g:Profiler terms. We report H2 as confirmed-then-explained, not as evidence that miRNA targets form network modules.

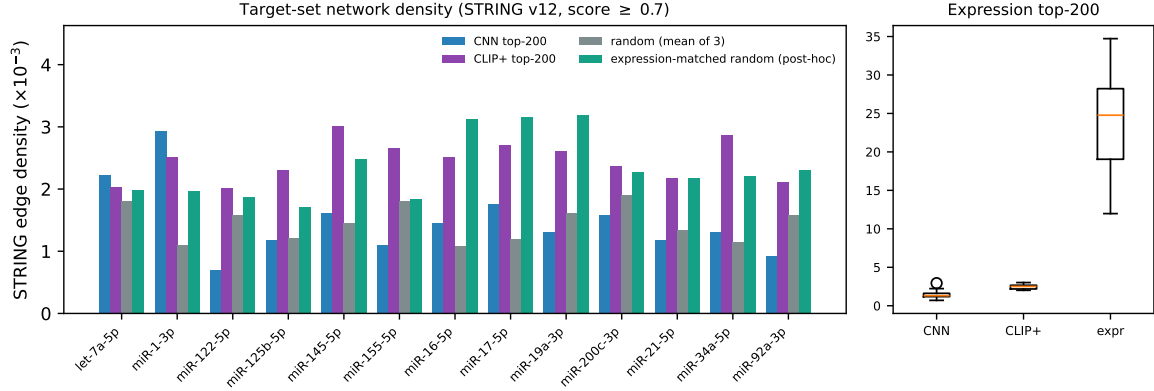


Figure 11: Left: STRING edge density per miRNA for CNN top-200, CLIP+ top-200, seeded random and expression-matched random sets. Right: distribution across miRNAs with the expression top-200 control, which is several-fold denser.

4.16 miRNA abundance does not predict which miRNAs the CNN ranks well (ENCODE; negative)

CLIP labels need AGO loaded with the miRNA, so abundant miRNAs could give cleaner labels and higher per-miRNA AUROC. We measured abundance from 21 ENCODE GRCh38 microRNA-seq quantification files covering 12 cell lines (accessions in `results/encode_mirna_abundance.json`; Ensembl REST lookup/id for gene symbols; `scripts/encode_mirna_abundance.py`; hermetic tests). Per file we take $a_g = \log_2(\text{CPM}_g + 1)$, sum precursor loci per mature miRNA (precursor counts cannot separate the 5p and 3p arms), average replicates and take the median across lines. Gates: every file has at least 1,500 gene rows; miR-21 sits in the panel's top decile (17.8 vs threshold 12.3); 131/131 panel miRNAs map to a precursor. The pre-registered tissue gate failed: we expected miR-122 to peak in HepG2, but it is zero in all 12 lines. We record that failure and do not retune it. A replacement control chosen after the failure (post-hoc, not pre-registered) passes: MIR142's top three lines are the three blood lines (GM12878, HL-60, K562), and MIR302A, MIR302B and MIR367 each peak in H1 embryonic stem cells. All three hypotheses are falsified on $n = 131$ miRNAs: abundance versus CNN AUROC $\rho = 0.076$ ($p = 0.195$), versus the CNN's gain over site-count covariates $\rho = 0.099$ ($p = 0.130$), and the partial correlation given $\log n_{pos}$ $\rho = 0.050$ ($p = 0.284$; Figure 12). Abundance is also unrelated to the covariate-only AUROC ($\rho = 0.010$). A caveat: ENCORI CLIP pools many cell types, and none of the 12 ENCODE lines is HEK293, so a cell-matched abundance could still matter.

4.17 Tissue target avoidance (GTEx): conserved targets avoid the miRNA's tissue, CNN non-conserved targets do not

Targets of a tissue-specific miRNA should be relatively depleted in that tissue. We test this with GTEx v8 gene median TPM (`scripts/gtex_target_avoidance.py`; `results/gtex_target_avoidance.json`; hermetic tests) on 8 pre-registered pairs (miR-1-3p, miR-133b, miR-206: skeletal muscle; miR-122-5p: liver; miR-9-5p, miR-128-3p: brain cortex; miR-223-3p: whole blood; miR-375: pancreas). Relative expression is $r_g(t) = \log_2(\text{TPM}_{g,t} + 1) - \text{median}_{t'} \log_2(\text{TPM}_{g,t'} + 1)$ and depletion is the Mann-Whitney probability $\text{AUC}_{dep} = P(r_{\text{set}} < r_{\text{rest}})$. Gates passed: ALB peaks in liver, ACTA1 in skeletal muscle, median mapping 95.6%. CNN top-200 genes are not depleted in the matched tissue (H1: 1 wins, 7 losses, $p = 0.996$).

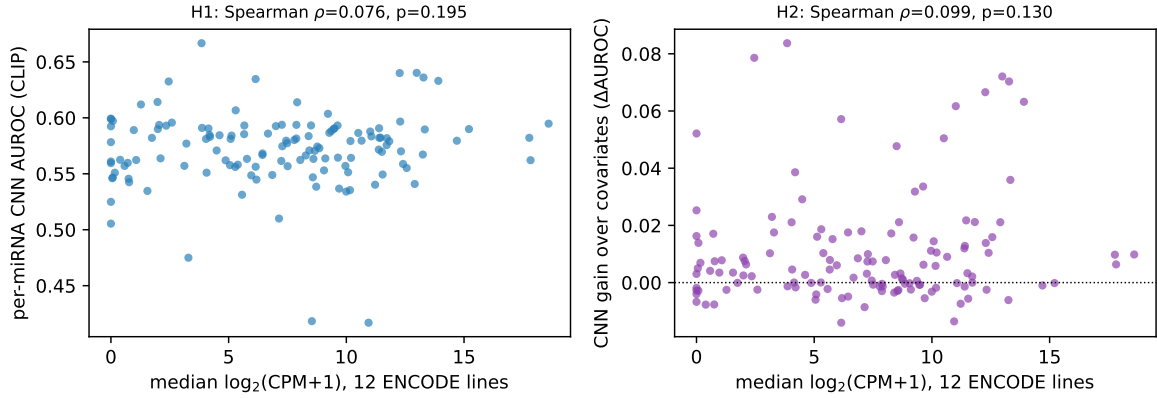


Figure 12: Per-miRNA ENCODE abundance (median over 12 cell lines) versus CNN AUROC on CLIP labels (left) and versus the CNN’s gain over site-count covariates (right).

and not more depleted there than in the other design tissues (H3: 5/3, $p = 0.363$); both are falsified. Pre-registered H2 (TargetScan genes inside the CLIP universe) turned out to be empty by construction, since that universe excludes every gene with a conserved site for the seed; we record it as not evaluable. A post-hoc reference arm that we did not pre-register uses all TargetScan 3’UTR genes as the universe and the top-200 conserved-site genes as the set: they are depleted in the matched tissue for 8/8 pairs ($p = 0.004$) and more so than in other tissues for 7/8 ($p = 0.035$; Figure 13), which recovers the classic avoidance signal (Farh et al. 2005). So the method sees avoidance where it is expected, and the CNN’s non-conserved candidates show none. Two caveats: the arms use different universes (CNN scores exist only for the CLIP universe), and conservation and scorer are confounded, so this does not separate “non-conserved sites are not functional in vivo” from “the CNN ranks them poorly.” Both readings are negative for the non-conserved discovery claim.

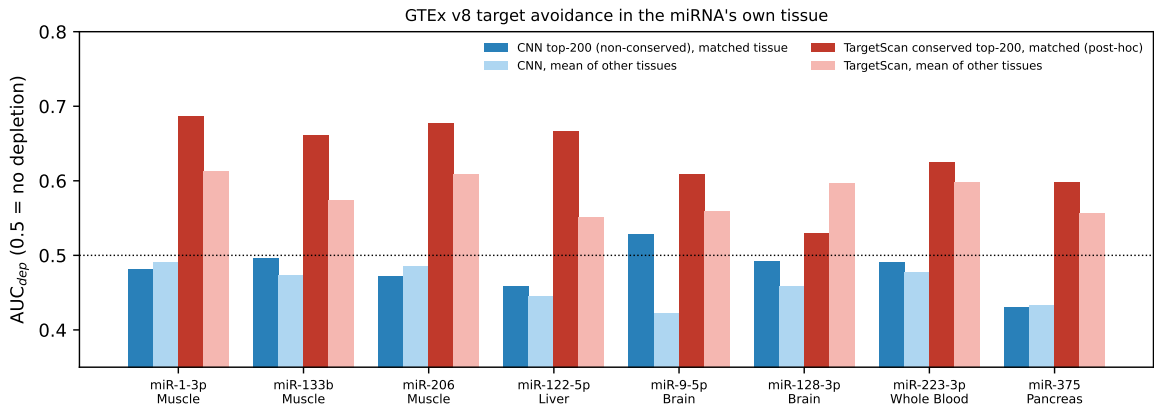


Figure 13: Depletion (AUC_{dep}) of target sets in each miRNA’s own GTEx tissue versus the mean over the other design tissues. Blue: CNN top-200 in the CLIP universe (pre-registered). Red: TargetScan conserved top-200 in the all-UTR universe (post-hoc reference).

4.18 Reverse lookup (Enrichr): CNN top-200 sets name their own miRNA beyond expression, but the site-count win does not replicate

A stricter use of a target list is to ask whether it identifies its miRNA among thousands. We submit each set to the Enrichr API against the miRTarBase_2017 library (3,240 miRNA terms; `scripts/enrichr_reverse_lookup.py`; `results/enrichr_reverse_lookup.json`; hermetic tests) and record the rank of the query miRNA’s own term by Enrichr’s Fisher p -value. Panel, sets and seeds match the g:Profiler section. Gates passed: the library’s own miR-21-5p set returns miR-21-5p at rank 1; the median random-set own-term rank is 283; all 13 miRNAs are library terms. H1 is confirmed: CNN top-200 sets rank their own miRNA better than random for 13/13 miRNAs ($p = 1.2 \times 10^{-4}$; median rank 3), and so do CLIP-positive sets (H2, 13/13; partly circular because miRTarBase includes CLIP-derived entries). H3 is falsified: the HEK293 expression top-200 control also ranks the own term well, and the CNN beats it on only 7/13 (4 losses, $p = 0.274$). Because the CNN’s univariate CLIP signal was earlier shown to be mostly site-count confounding, and expression dominates the other audits, we added two controls after H1 came back positive (post-hoc, not pre-registered). The CNN beats a site-count top-200 set drawn from the same universe (11 wins, 0 losses, 2 ties, $p = 4.9 \times 10^{-4}$) and random sets matched to the CNN set’s expression-decile histogram (13 wins, 0 losses, $p = 1.2 \times 10^{-4}$; Figure 14). Limits: miRTarBase 2017 overlaps the miRTarBase 10.0 labels used in the earlier benchmark, so this is a second view of the same kind of evidence; heavily studied miRNAs (miR-21, miR-155) reach rank 1 even from random sets; and both controls were post-hoc. We therefore pre-registered a held-out replication (`scripts/enrichr_replication.py`, committed before its results; `results/enrichr_replication.json`) on 30 new miRNAs chosen by a seeded shuffle, with both controls fixed in advance. Median own-term ranks were CNN 10.5, site count 70, expression-matched random 433 and uniform random 506. The CNN beats uniform random (26/4, $p = 3.0 \times 10^{-5}$) and expression-matched random (25/5, $p = 1.6 \times 10^{-4}$), but not site count (16 wins, 10 losses, $p = 0.163$). By the pre-registered rule the discovery claim does not replicate: a CNN target list names its miRNA far better than chance and than expression, but most of that comes from seed-site count, as in the per-miRNA feature audit. The site-count win on the discovery panel was a small-panel result.

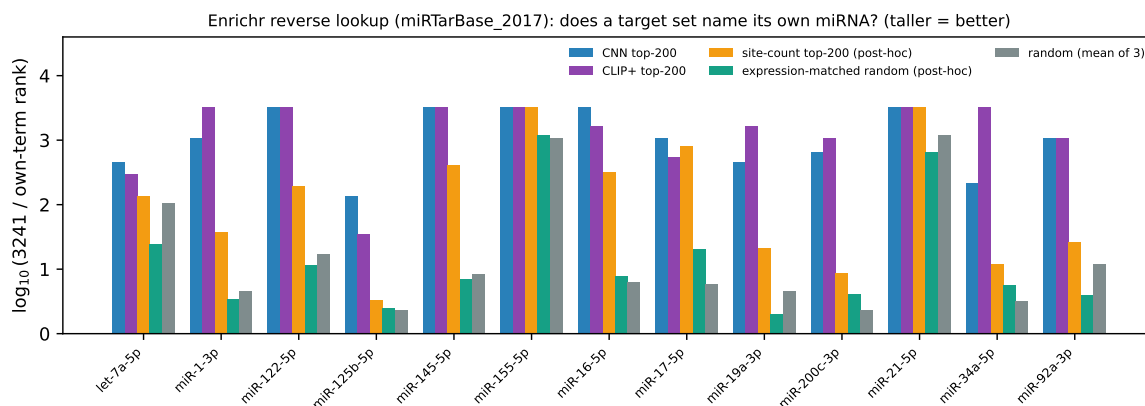


Figure 14: Rank of each miRNA’s own miRTarBase_2017 term when its target set is submitted to Enrichr plotted as $\log_{10}(3241/\text{rank})$, so taller bars mean better ranks. Site-count and expression-matched arms are post-hoc controls.

4.19 Reactome pathway audit with controls fixed in advance: two gates failed, verdicts inconclusive

We repeated the pathway question with Reactome v97 (AnalysisService projection; `scripts/reactome_enrichment.py`; `results/reactome_enrichment.json`; hermetic tests), this time with the site-count and expression-matched controls pre-registered. Per miRNA we count pathways at entities FDR < 0.05 for CNN top-200, site-count top-200, three expression-matched and three uniform random sets. Two of three gates failed as written. G1 required a pathway named “Cell Cycle” among the top three for a 20-gene cell-cycle control; the control returned 205 significant pathways, but the top three are tied at the same FDR and none has that name (a post-hoc look finds 7 such pathways lower in the list). G3 required fewer than 10% unmapped genes; the median is 39.8%, because Reactome curates only part of the genome. Both are design errors on our side, and we do not rewrite the gates. G2 passed (median uniform-set count 0). Under failed gates we report the numbers but call the audit inconclusive: CNN versus uniform random 3/6 ($p = 0.910$), versus site count 2/2 ($p = 0.688$), versus expression-matched 4/6 ($p = 0.828$). Most sets in every arm have zero significant pathways, with a few large random-set outliers (Figure 15). None of this points toward pathway coherence, which agrees with the g:Profiler result.

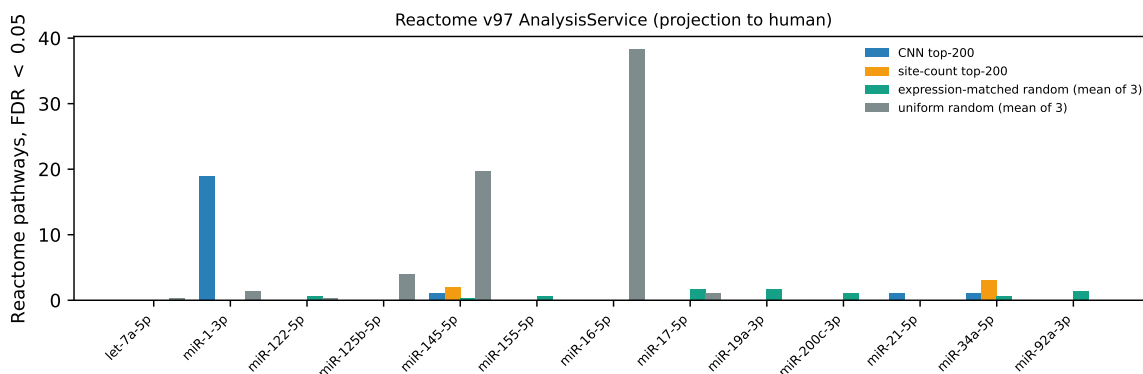


Figure 15: Significant Reactome pathways per miRNA for four arms of 200 genes. Gates G1 and G3 failed, so these counts are descriptive.

4.20 Input integrity: every miRNA sequence and seed used here matches RNAcentral

All scorers in this paper key on the miRNA seed (nt 2–8) taken from TargetScan’s `miR_Family_Info`. A wrong sequence would silently corrupt every benchmark, so we checked each miRNA used in the CLIP and miRTarBase analyses against RNAcentral by miRBase accession via the RNAcentral REST API (script and results: `rnacentral_seq_audit`; hermetic tests). Both hypotheses are confirmed (Table 1). One transport note: for a few accessions the endpoint served an HTML page to Python requests but JSON to curl, so the fetcher falls back to curl when the body is not JSON; an earlier run without the fallback left one or two accessions unresolved and was discarded (it was also overwritten by a duplicate process). No analysis choice changed. This rules out sequence errors as an explanation for the negatives above.

Quantity	Value
miRNAs used / with MIMAT accession	132 / 132
Resolved in RNAcentral	132 (100.0%)
G1: let-7a-5p exact sequence	pass
G3: seed identity under deranged map (control)	0.8%
H1: seed identity, resolved miRNAs	100.0%
H2: full-length identity, resolved miRNAs	100.0%

Table 1: RNAcentral sequence audit. The derangement control shows the identity test can fail.

4.21 Independent curated labels (OmniPath, miRTarBase removed): CNN above chance, but not above site count or expression

The miRTarBase and Enrichr analyses share one curation source. OmniPath aggregates several (script and results: `omnipath_independent`; hermetic tests), so we kept only interactions whose sources exclude miRTarBase (2,795 of 11,240; ncRDeathDB, miRecords, miR2Disease, miRDeathDB, SIGNOR) and scored them inside each miRNA’s CLIP universe of non-conserved seed-match genes. The site-count, expression and label-permutation controls were fixed before scoring. Gates passed: 13 miRNAs have at least five positives (bar 10), and the median permutation-mean CNN AUROC is 0.504. The CNN is above chance (H1: 11/2, $p = 0.011$, median AUROC 0.566), but it does not beat site count (H2: 8/5, $p = 0.291$, median 0.572) and loses to HEK293 expression (H3: 3/10, median 0.689; Figure 16). This matches the Enrichr replication on a label set that shares no miRTarBase rows: curated low-throughput targets favour expressed genes, and whatever the CNN adds over counting seed sites is not detectable with 5–29 positives per miRNA.

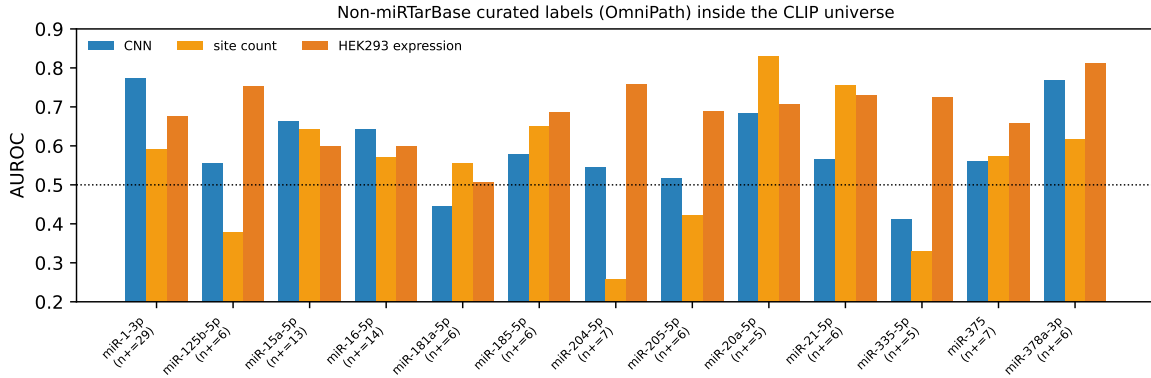


Figure 16: Per-miRNA AUROC on OmniPath interactions with miRTarBase-sourced rows removed, for the CNN, site count and HEK293 expression.

4.22 Transcription-regulator enrichment (UniProt): present in conserved targets, absent in CNN non-conserved targets

Conserved miRNA targets are known to be enriched for transcription regulators. We test whether CNN non-conserved targets share this property using UniProtKB release 2026_03 (reviewed human entries with keyword KW-0805, 2,374 of 20,190; script and results: `uniprot_tf_enrichment`; hermetic tests). The statistic is the fraction of a 200-gene set carrying KW-0805; site-count and expression-matched controls were fixed in advance. Gates passed. The positive control recovers the classic result: TargetScan conserved

top-200 sets are richer in transcription regulators than random UTR genes for 13/13 miRNAs ($p = 1.2 \times 10^{-4}$), and KW-0805 prevalence is 11.8%. CNN top-200 sets show no enrichment: versus uniform random 6/7 ($p = 0.709$), versus site count 4/9 ($p = 0.954$), versus expression-matched 7/6 ($p = 0.500$); all three are falsified (Figure 17). Together with the GTEx avoidance result this is the second functional signature that conserved targets carry and CNN non-conserved candidates lack. The same caveat applies: conservation and scorer are confounded, and the universes differ between arms.

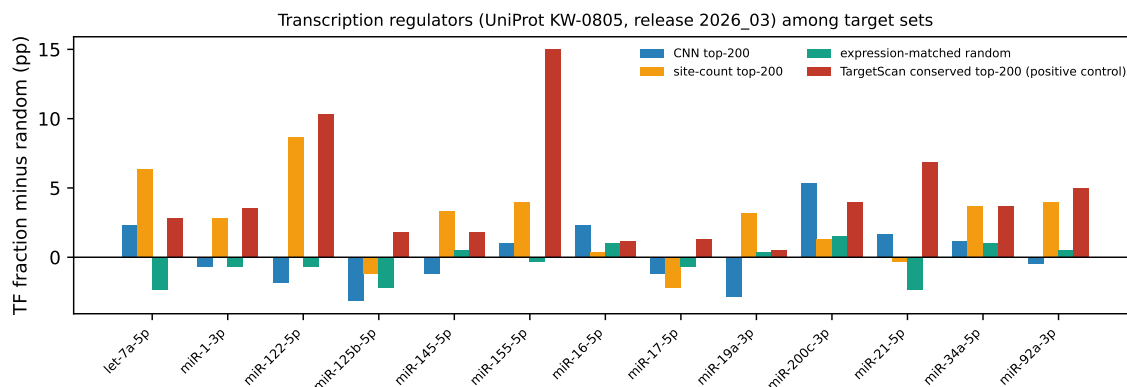


Figure 17: Transcription-regulator fraction of each 200-gene set minus its random baseline (percentage points). Red: TargetScan conserved targets against random UTR genes (positive control).

4.23 Loss-of-function constraint (gnomAD): conserved targets are constrained, CNN candidates are the least constrained set

Conserved miRNA targets tend to be dosage-sensitive. We compared the median LOEUF (gnomAD v2.1.1 upper bound of observed/expected loss-of-function; lower means more constrained; 19,155 genes; script and results: `gnomad.constraint`) of 200-gene sets, with the same controls fixed in advance. Gates passed: 97.6% of CLIP-universe genes have a LOEUF, and TargetScan conserved top-200 sets are more constrained than random UTR genes for 13/13 miRNAs ($p = 1.2 \times 10^{-4}$). All three pre-registered CNN hypotheses are falsified, and in the opposite direction: the CNN top-200 has a higher median LOEUF than uniform random (13/13), site count (13/13) and expression-matched sets (13/13; Figure 18). Not pre-registered: the two-sided sign test of this reversal gives $p = 2.4 \times 10^{-4}$ against uniform and $p = 2.4 \times 10^{-4}$ against expression-matched sets. Site-count sets are more constrained than uniform random for 10/13 miRNAs, as expected if they favour genes with long 3' UTRs. The CNN's preference for genes with few, strong sites may select short-UTR, less constrained genes. This post-hoc finding needs a held-out, UTR-length-matched replication before it can be read as biology, and we make no claim beyond the description.

Pre-registered held-out replication. We committed a replication script before running it (`gnomad.replication`): 30 CLIP miRNAs outside the discovery panel, a one-sided test in the discovered direction, and a new UTR-length-matched control (decile matching on the longest 3' UTR). Gates passed (97.8% coverage). The reversal replicates: the CNN top-200 is less constrained than uniform random in 30/30 miRNAs ($p = 9.3 \times 10^{-10}$; median LOEUF difference +0.192), than expression-matched sets in 30/30 ($p = 9.3 \times 10^{-10}$; +0.161), and than UTR-length-matched sets in 25/30 ($p = 1.6 \times 10^{-4}$; +0.084; Figure 19). CNN top-200 genes have shorter UTRs than their universe in 30/30 miRNAs, and length matching shrinks the gap, but it does not

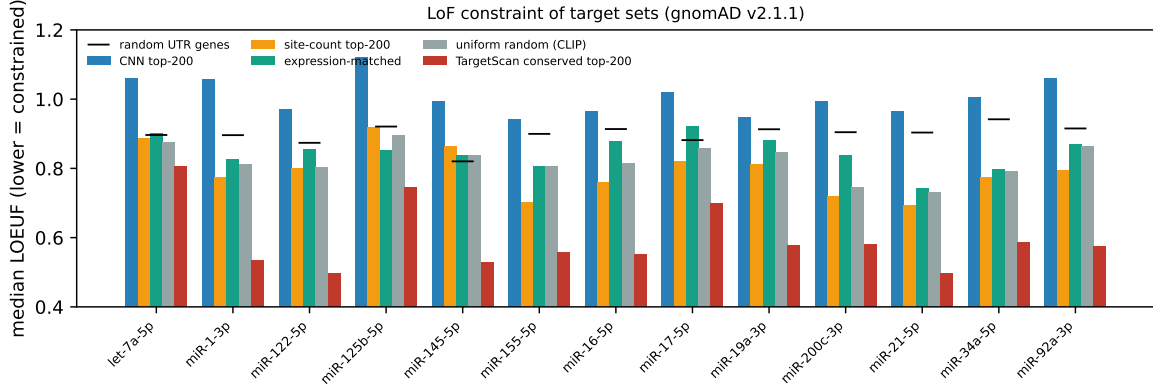


Figure 18: Median LOEUF of each 200-gene set. Black dashes: random UTR genes, the comparator for the TargetScan positive control.

remove it. We name this as a falsifiable finding: *among non-conserved seed-match genes, the CNN's top-ranked candidates are depleted of loss-of-function-constrained genes, beyond what HEK293 expression or 3' UTR length explain.* It would be falsified by (i) a finer length control (for example exact length bins or site-density matching) that removes the gap, or (ii) the same depletion appearing for a random-weight CNN, which would show it comes from the input encoding rather than the learned model. Neither test has been run. One reading is purifying selection against strong non-conserved sites in dosage-sensitive genes, but we have not tested it and do not claim it.

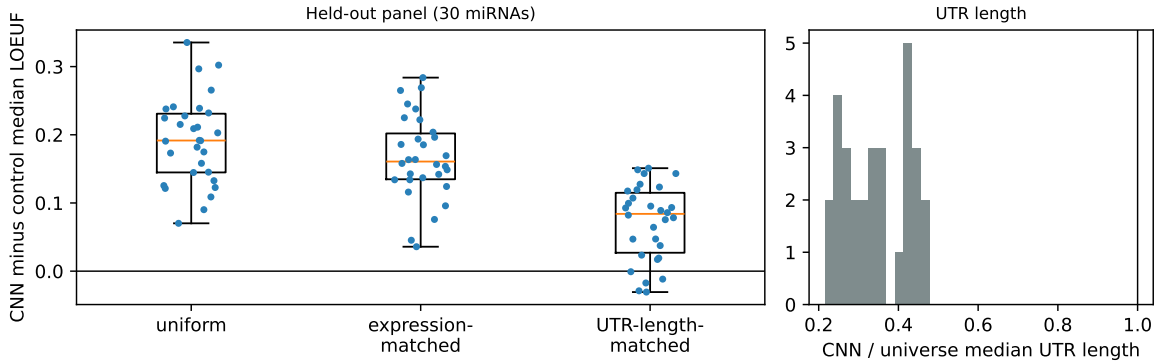


Figure 19: Pre-registered replication on 30 held-out miRNAs. Left: CNN top-200 median LOEUF minus each control (positive = CNN less constrained). Right: CNN top-200 median UTR length relative to its universe.

4.24 Independent dosage label (ClinGen): the depletion beyond UTR length does not replicate

gnomAD LOEUF is a population statistic, so we tested the named finding against an independent label: expert-curated ClinGen haploinsufficient genes (score 3, 420 genes), on all 43 miRNAs (13 discovery + 30 held-out). The script was committed before it was run (`clingen_dosage`). Gates passed: the pooled CLIP universe holds 394 HI genes, and TargetScan conserved top-200 sets hold more HI genes than random

UTR genes (32/5 miRNAs, $p = 3.7 \times 10^{-6}$; pooled 307 vs 166). The CNN sets hold fewer HI genes than

Pre-registered test	wins/losses	p	pooled CNN / control	verdict
C1 CNN < uniform	26/16	0.082	158 / 205.3	FALSIFIED
C2 CNN < expression-matched	28/14	0.022	158 / 191.7	CONFIRMED
C3 CNN < UTR-length-matched	24/16	0.134	158 / 159.3	FALSIFIED

Table 2: ClinGen HI genes in CNN top-200 sets. Per-miRNA sign tests (ties excluded) and pooled counts over 43 miRNAs.

expression-matched sets (C2 confirmed; pooled ratio 0.82), but the depletion against uniform sets is not significant (C1) and it vanishes against UTR-length-matched sets (pooled ratio 0.99; C3 falsified; Figure 20). On this independent label, UTR length explains the depletion. The gnomAD finding therefore survives only in its weaker form (CNN candidates are less constrained than expression-matched genes); its “beyond UTR length” clause is supported by LOEUF but not by ClinGen. HI genes are sparse (about 4 per set), so C3 has little power; we report the disagreement rather than choosing between the labels.

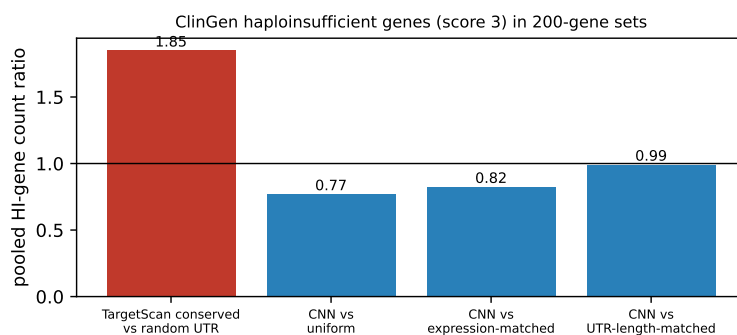


Figure 20: Pooled HI-gene count in each set divided by its control. Red: positive control.

5 Documented negatives

(1) Sequence-only DuplexCNN does not beat ridge. (2) CLIP effect size is small. (3) miR-125b-5p and miR-7-5p per-miRNA AUROCs (negated convention) are at/below chance — reported per miRNA, not averaged away. (4) Pair-map branch contributes only $\Delta r = +0.0029$ (ablation). (5) miRDB v6.0 beats our CNN on miRTarBase rows (pooled AUROC 0.787 vs 0.685, 74/84 miRNAs, $p = 2.4 \times 10^{-12}$). (6) On TarBase v9.0 labels the CNN loses to miRDB on 73/83 miRNAs and to the plain seed-site count per miRNA (CNN better on only 32/83, $p = 0.0078$). (7) HEK293 expression alone (HPA nTPM, AUROC 0.682) beats all sequence models on CLIP labels and cuts the CNN’s conditional pooled gain from +0.0048 to +0.0008. (8) g:Profiler finds no pathway coherence in CNN top-200 (13/13 miRNAs zero terms) or CLIP top-200 target sets; an expression top-200 control gives 47–115 terms. (9) STRING network density: CNN top-200 no denser than random; the CLIP+ excess over random (13/13) does not survive an expression-matched control (8 wins, 4 losses, $p = 0.19$). (10) ENCODE miRNA abundance (12 lines) does not predict per-miRNA CNN AUROC ($p = 0.076$, $p = 0.195$) or gain; pre-registered miR-122/HepG2 gate failed (zero in all lines). (11) GTEx: CNN non-conserved top-200 targets show no tissue avoidance (1 win, 7 losses) and no tissue specificity (5/3, $p = 0.36$); pre-registered TargetScan arm empty by construction. (12) Enrichr: HEK293 expression top-200 names the right miRNA nearly as well as the CNN (H3 falsified, 7/4); the CNN’s reverse-lookup edge over site count (11/0 on the discovery panel) failed pre-registered held-out replication (16/10,

$p = 0.16$, 30 miRNAs). (13) Reactome audit: pre-registered gates G1 (tie-ordered top-3 rule) and G3 (40% unmapped) failed; verdicts inconclusive, no sign of pathway coherence. (14) OmniPath non-miRTarBase labels: CNN above chance (11/2) but not above site count (8/5, $p = 0.29$) and below HEK293 expression (3/10). (15) UniProt KW-0805: conserved targets enriched for transcription regulators (13/0), CNN non-conserved targets not (6/7 vs random, 4/9 vs site count). (16) gnomAD LOEUF: conserved targets constrained (13/0); CNN top-200 less constrained than every control (0/13). The reversal was found post hoc; its pre-registered held-out replication (30 miRNAs) confirmed it against uniform (30/0), expression-matched (30/0) and UTR-length-matched (25/5) sets. The finer controls are still to be run. (17) ClinGen HI genes (independent label, pre-registered, 43 miRNAs): the CNN depletion holds against expression-matched sets (28/14) but not against uniform (26/16, $p = 0.082$) or UTR-length-matched sets (pooled ratio 0.99). The gnomAD clause *beyond UTR length* is not supported by ClinGen.

6 Falsification paths

miRTarBase 10.0 and TarBase v9.0 benchmarks are now done (see above); wet-lab: luciferase reporters for the top 10 diverse pairs; extended-panel CLIP rerun over the 132-miRNA ENCORI panel completed (delta +0.0060, $p=0.048$); remaining: library-matched CLIP/RNA-seq normalization.

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7 Mathematical derivations

Notation matches `src/duplexcnn` and `scripts/clip_falsify.py`; every statistic below appears in a committed results JSON.

7.1 Convolutional duplex scoring

The pair-map P is scored by stacked convolutions. One layer with kernel $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k}$ and bias b :

$$h_{c,j}^{(\ell+1)} = \sigma \left(\sum_{c'} \sum_{u=0}^{k-1} W_{c,c',u} h_{c',j+u-p}^{(\ell)} + b_c \right), \quad \sigma(x) = \max(0, x), \quad (3)$$

with padding $p = \lfloor k/2 \rfloor$ keeping length fixed so positional pairing information survives to the head. Training minimizes MSE against the TargetScan context++ score; the held-out unit is the *gene*, so no transcript can contribute to both fit and evaluation.

7.2 Pearson and Spearman benchmarks

Held-out agreement with context++ is

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2 \sum_i (\hat{y}_i - \bar{\hat{y}})^2}}, \quad r_s = r(\text{rank}(y), \text{rank}(\hat{y})), \quad (4)$$

the second being Pearson on ranks, robust to monotone distortion of the score scale. DuplexCNN: $r = 0.8121$, $r_s = 0.7928$ (committed ablation JSON).

7.3 Fisher exact test for CLIP enrichment

For one miRNA, genes partition by CNN min-score (high/low) and CLIP support (supported/not) into a 2×2 table with margins fixed by construction. Under the null of independence the count x of high-scoring CLIP-supported genes is hypergeometric:

$$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad p = \sum_{x' \geq x} P(X = x'), \quad (5)$$

an exact small-sample test that needs no asymptotic approximation.

7.4 Odds ratio

Effect size is reported as

$$\text{OR} = \frac{x/(n-x)}{(K-x)/(N-K-n+x)}, \quad (6)$$

so a per-miRNA AUROC of 0.55 can be read directly as an enrichment strength rather than a bare significance flag.

7.5 AUROC as a Mann–Whitney statistic

The per-miRNA CNN-min AUROC against CLIP labels is

$$\text{AUROC} = \frac{1}{n_+ n_-} \sum_{i \in +} \sum_{j \in -} \left(\mathbf{1}[s_i > s_j] + \frac{1}{2} \mathbf{1}[s_i = s_j] \right), \quad (7)$$

i.e. the probability a random CLIP-supported gene outscores a random unsupported one; chance is 0.5 by construction.

7.6 Bootstrap confidence interval for ΔAUROC

The base-vs-base+CNN comparison resamples (miRNA, gene) units with replacement $B = 1000$ times, refits nothing (scores are fixed), and recomputes the AUROC gap:

$$\Delta_b = \text{AUROC}_b^{\text{base+CNN}} - \text{AUROC}_b^{\text{base}}, \quad \text{CI}_{95} = [\hat{Q}_\Delta(0.025), \hat{Q}_\Delta(0.975)], \quad (8)$$

which yielded $[+0.0057, +0.0093]$ on the 23-miRNA panel — the interval excludes zero, so the sign of the effect is not sampling noise.

7.7 Permutation null

To test whether the CNN adds signal beyond covariates within miRNA, labels are shuffled *inside* each miRNA stratum $B = 20$ times:

$$p = \frac{1 + \#\{b : \Delta_b \geq \Delta_{\text{obs}}\}}{B + 1}, \quad (9)$$

preserving miRNA-level base rates under the null. The 23-miRNA panel gave $p = 0.048$; the 136-miRNA rerun (this paper’s extended falsification) uses the identical null so the two numbers are commensurable.

7.8 Seed-match enumerator

The discovery scan counts exact seed complementarity. For UTR u and miRNA m with seed σ_m (positions 2–8),

$$n_{8\text{mer}}(u, m) = \sum_j \mathbf{1}[u_{j:j+8} = \text{revcomp}(\sigma_m A1)], \quad n_{\text{total}} = \sum_m \sum_u (n_{8\text{mer}} + n_{7\text{mer-A1}}), \quad (10)$$

enumerating 1,453,439 novel non-conserved matches — the scale that makes a learned prioritizer necessary and the falsification non-trivial.

7.9 Wilcoxon signed-rank for paired per-gene comparisons

Where designs are compared per gene g (full vs ablated model), differences d_g are ranked by magnitude and

$$W = \sum_{g:d_g > 0} \text{rank}(|d_g|), \quad p \text{ from the exact null distribution of } W, \quad (11)$$

a distribution-free paired test whose $W = 0$ (all differences one direction) is the strongest possible ordinal outcome at a given n .